DANIEL BENJAMIN

contact@danielrbenjamin.com | 236-562-2566 | linkedin.com/in/danielrbenjamin | danielrbenjamin.com
available starting May 2026 for a 4-month or 8-month co-op position

EDUCATION

University of British Columbia Expected Graduation: May 2028
Bachelor of Applied Science – Mechanical Engineering

SUMMARY

Third-year Mechanical Engineering student with hands-on experience in manufacturing and prototyping, supported by skills in CAD, simulation, and building systems design. Proficient in SOLIDWORKS, ANSYS, AutoCAD, and Excel, and experienced in developing tools using Python, PowerShell, and AutoLISP. Combines practical shop experience with strong project coordination and documentation in multidisciplinary teams.

SELECTED EXPERIENCE

Chassis Team Lead (prev. Team Member)**,** UBC Formula Electric Sep 2023 – Present

- Led 7-person chassis subteam using agile project management practices, tracking tasks and milestones using Jira and Gantt charts to keep the 2026 competition vehicle on schedule
- Used SOLIDWORKS to design an FSAE chassis frame and developed Python scripts to calculate forces under various braking and cornering load cases for torsional rigidity simulations in ANSYS Mechanical
- Designed and validated multiple waterproof 3D-printed enclosures for circuit boards against IPX5 standards
- Planned and executed carbon fiber + glass fiber composite layups for car body panels and aero kit
- Implemented a CAD document control system using SVN on Oracle Cloud to track and manage design revisions
- Designed a custom 3D printer management dashboard and Slack integration to monitor print status and automate notifications, manage reservations and reduce downtime of 3D printers
- Wrote detailed design documentation and SOPs to ensure knowledge transfer for future team members

Mechanical Co-op, Avalon Mechanical May 2025 – Aug 2025

- Developed AutoLISP automation scripts reducing CAD drafting time by 7+ hours per week
- Produced HVAC, plumbing, and fire suppression layouts in AutoCAD for commercial and residential projects
- Performed hallway pressurization analysis and fan sizing using Excel and TRACE, calculating required airflow and pressure differentials to meet NFPA and provincial code requirements
- Validated fire sprinkler hydraulic systems using Elite Software FIRE for code-compliant designs

Undergraduate Research Assistant, UBC MEMS Lab May 2024 – Feb 2025

- Systematically reviewed 15+ prosthetic hand designs to identify common successful design elements
- Rapidly prototyped design iterations using a custom-built multi-extrusion 3D printer running RepRapFirmware
- Designed a robotic hand driven by soft pneumatic actuators 25% better at grasping than prior iterations
- Helped develop a demonstration platform to mimic human hand gestures using MATLAB and C

Volunteer, Victoria Hand Project July 2022 – Present

- Assembled and repaired 3D-printed low-cost prosthetic arms with voluntary open/close functionality
- Identified process improvements and contributed new assembly documentation to improve quality

SVNWorks, Personal Project danielrbenjamin.com/projects/svnworks

- Built a custom app to integrate between SOLIDWORKS, File Explorer, and an SVN server to streamline a check-in/check-out style version-controlled document management system for a student engineering team

AI-Driven Predictive Maintenance System, Course Project github.com/danielrbenjamin/manu465-project

- Applied ML in Python to predict maintenance and detect anomalies in smart manufacturing equipment

WeBWorKer Chrome Extension, Personal Project github.com/danielrbenjamin/WeBWorKer

- Published Chrome extension reaching 900+ users, enhancing homework platform UX using JS and HTML/CSS

SKILLS

Mechanical: SOLIDWORKS, AutoCAD+AutoLISP, ANSYS, Composites, Machining, 3D Printing, FMEA
Software: MS Office (Excel, Project), C, MATLAB, Python, JS, Git, SVN, Jira, Bluebeam, Docker, Adobe CC